

MATH 213

Randomness and inferences

Big ideas when learning from data

Causation: We often gather data to learn about the effects of actions such as taking a certain medicine or following a certain diet. How do we gather and use data to understand the effects of actions?

Generalization: We usually only gather data about a sample, a small part of the bigger population that we want to know about. How can we draw conclusions about the population using what we learn about the sample?

Principles from earlier this semester

- It is reasonable to draw cause/effect conclusions from a randomized experiment
- It is reasonable to generalize from a random sample to a population.

Going deeper

For the past few days, we considered a very specific hypothesis (theory) about the data

- In a randomized experiment: treatment and control would have exactly the same effect on every case in the sample. Difference between groups is from randomization alone.
- Other situations: a skeptic's model, along the lines of “individuals in the study were guessing.” Study results reflect guessing alone.

We evaluated these theories (null hypotheses) using simulation of the random process, either physically (with coin flipping) or on a computer (with R or an applet).

- How likely is the random process to create data like what we saw?
- If unlikely, we decided that null hypothesis theory was not plausible.

A new approach

In warm-up activity:

- **Random** sample of 765 adults, where 322 (42%) could not cover \$400.
- The 42% is a *statistic*.
- *Parameter*: percentage of population that could not cover \$400.
- First guess (Chapter 1): 42% of population could not cover \$400.
- But sample is probably not a perfect representation of population.
- How precisely can we estimate parameter? Exactly 42%? Between 40% and 44%? Between 30% and 50%?
- We will learn some ways to answer that question